IP Address – Complete Notes

Decoding IP Addresses | Topic-wise Explanation with Diagrams

Is PDF me ye topics cover kiye gaye hain:

1 IP Address kya hota hai?

IP Address (Internet Protocol Address) ek unique numeric address hota hai jo internet ya kisi bhi network se connected har device ko diya jata hai. Jis tarah har ghar ka ek unique postal address hota hai jisse dak (parcel/letter) sahi jagah pahunchti hai, usi tarah IP address ek device ki "digital pehchaan" hai jisse data (packets) sahi device tak pahunchta hai.

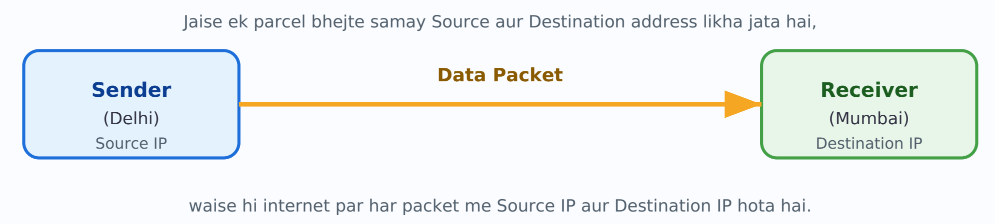


Fig 1.1 – IP Address parcel/courier address ki tarah kaam karta hai

Key Points

- Har device (mobile, laptop, server, router) ko internet se judne ke liye ek IP Address chahiye.
- IP Address **unique** hota hai — koi bhi do device same network par same IP address nahi rakh sakte.
- Ye Internet Protocol (IP) ke rules ke through assign hota hai.

2 IP Address ka Structure (32-bit / 4 Bytes)

IPv4 Address **32 bits** ka hota hai, jise **4 parts (octets)** me divide kiya jata hai. Har part **8 bits (1 Byte)** ka hota hai. Isliye total = 4 × 8 = 32 bits = 4 Bytes.

Example: 192.36.58.24

192	36	58	24
Octet 1 (8 bit)	Octet 2 (8 bit)	Octet 3 (8 bit)	Octet 4 (8 bit)

Total = 8+8+8+8 = 32 bits = 4 Bytes

Fig 2.1 – IPv4 Address ke 4 Octets (Dotted Decimal Notation)

Har Octet ki Range 0–255 kyun hoti hai?

Har octet 8 bits ka hota hai. 8-bit binary number se jitni bhi combinations ban sakti hain wo 2^8 = **256** hoti hain. Binary counting `0` se shuru hoti hai, isliye range **0 se 255** tak hoti hai (total 256 values).

Example: Binary `00000000` = Decimal 0, aur Binary `11111111` = Decimal 255. Isliye har octet minimum 0 aur maximum 255 ho sakta hai.

3 IPv4 vs IPv6

Jaise-jaise internet users badhte gaye, IPv4 ke addresses kam padne lage (kyunki IPv4 me limited unique addresses hi ban sakte hain). Isi problem ko solve karne ke liye **IPv6** banaya gaya.

Feature	IPv4	IPv6
Bit Size	32-bit	128-bit
Format	Decimal (e.g. 192.36.58.24)	Hexadecimal (e.g. 2001:0db8:...)
Total Bytes	4 Bytes	16 Bytes
Address Capacity	~4.3 Billion addresses	Bahut zyada (practically unlimited)
Notation	Dot (.) se separate	Colon (:) se separate

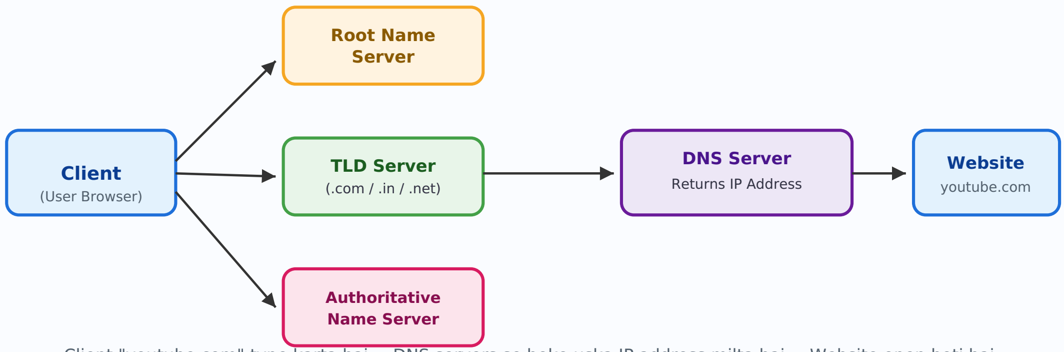
Yaad rakhein: IPv4 = 32 bit, IPv6 = 128 bit. IPv6 isliye laya gaya kyunki duniya bhar ke devices ke liye IPv4 ke addresses insufficient ho rahe the.

4 DNS – Domain Name System

Har website ka ek IP address hota hai (jaise YouTube ka koi IP address). Lekin IP address (jaise `192.36.58.24`) yaad rakhna mushkil hai. Isliye humne easy-to-remember **Domain Names** banaye (jaise `youtube.com`, `google.com`). **DNS (Domain Name System)** ek aisa system hai jo domain name ko uske corresponding IP address me convert karta hai.

DNS ko "Internet ki Phonebook" kaha jata hai

Jis tarah phonebook me naam se number dhoondte hain, DNS bhi domain name se IP address dhoondta hai.



Client "youtube.com" type karta hai → DNS servers se hoke uska IP address milta hai → Website open hoti hai

Fig 4.1 – DNS Resolution Process (Domain Name → IP Address)

DNS Resolution ke Steps

1. **Client Request:** User browser me `youtube.com` type karta hai.
2. **Root Name Server:** Query pehle Root server ke paas jaati hai, jo bataata hai ki `.com` domains kaha handle hote hain.
3. **TLD Server (Top Level Domain):** `.com`, `.in`, `.net` jaise extensions ke liye specific server query ko aage bhejta hai.
4. **Authoritative Name Server:** Ye server exact IP address batata hai us domain ka.
5. **Response:** IP address client ko wapas milta hai aur browser us IP address par connect karke website load karta hai.

Domain Name Extensions: `.com` (commercial), `.net` (network), `.in` (India), `.org` (organization) — ye sab TLD (Top Level Domain) ke examples hain.

5 Static IP vs Dynamic IP

Static IP Address	Dynamic IP Address
Address fix rehta hai, change nahi hota	Address time-to-time change hota rehta hai
Manually assign kiya jata hai	Automatically assign hota hai (DHCP se)
Servers, websites ke liye use hota hai	Home users/mobile devices ke liye common hai
Costly aur maintain karna padta hai	Cost-effective aur flexible

Example: Ek website server ka IP address usually **Static** hota hai taaki users hamesha usi address par site access kar sakein. Lekin aapke ghar ka WiFi router ka IP address ISP dwara **Dynamic** tarike se allot kiya jata hai — jo baar-baar change ho sakta hai.

6 IP Address Classes (A, B, C, D, E)

IPv4 addresses ko unke first octet ki value ke basis par 5 classes me divide kiya gaya hai. Ye classification network ka size decide karne me help karta hai.

Class	First Octet Range	Use
Class A	0 – 127	Bahut bade networks (large organizations)
Class B	128 – 191	Medium size networks
Class C	192 – 223	Chote networks (home/office)
Class D	224 – 239	Multicasting ke liye
Class E	240 – 255	Research/Experimental use

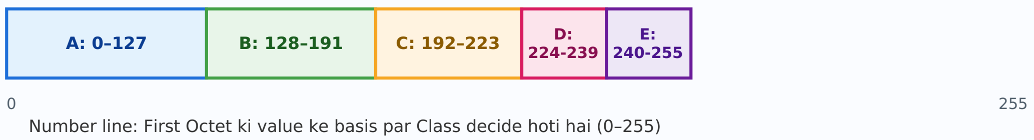


Fig 6.1 – IP Address Classes ka Number-Line Distribution

Yaad rakhne ka trick: Class A → Bade Network (large orgs), Class B → Medium Network, Class C → Chote Network (LAN/home), Class D → Multicast (group communication), Class E → Reserved/Experimental.

7 Network ID aur Host ID

Har IP Address do parts me divide hota hai:

- **Network ID:** Ye batata hai ki device kis network se belong karta hai.
- **Host ID:** Ye us network ke andar specific device (host) ko identify karta hai.

Example IP: 192 . 168 . 1 . 25

192 . 168 . 1	**25**
Network ID	Host ID

Fig 7.1 – IP Address ke andar Network ID aur Host ID ka split

Simple Analogy: Network ID ko "Colony/Society ka naam" samjho aur Host ID ko us colony ke andar "House Number" samjho. Colony same rehte hue bhi har ghar ka apna alag number hota hai.

Notes prepared from "Decoding IP Addresses" lecture — Topic-wise summary with diagrams.